

Predicting Perceived Interaction in Dancing Dyads: A Machine Learning Approach

Introduction

Human beings have the remarkable ability to synchronize their movements with musical stimuli. When dancing with a partner, individuals can coordinate their movements in response to both visual cues from their partner and the underlying rhythm, exemplifying a natural instance of intricate rhythmic-social entertainment. Interpersonal coordination is a well-studied topic in social psychology and has recently gained attention in the context of naturalistic movements in dyadic dancing [1, 3, 2]. Two perceptual variables are utilized to study interpersonal coordination, that is, similarity & interaction. The extent to which dancers horizontally orient towards each other was found to be an important correlate of perceived interaction [1, 3]. In the dyads that were maximally oriented towards each other, postural features like volumetric matching and gestural features like imitation and synchrony were identified to be associated with perceived interaction[2]. The current body of literature has identified features that exhibit only moderate correlations with perceived interaction, but no work exists that attempts to predict perceived interaction using machine learning. Studies have shown the importance of hand gestures in spoken communication [4]. [1] found that dancing dyads who move their hands faster tend to be perceived as more interactive and vice versa. Our study also explores the importance of synchronization between various joints of the two dancers in predicting interaction. We propose the following hypotheses:

- **H1:** The energy of dyads is positively correlated with interaction.
- **H2:** Dancers dancing with similar energy in a dyad are perceived as more interactive. In other words, the absolute value of the energy difference between two dancers correlates negatively with interaction.

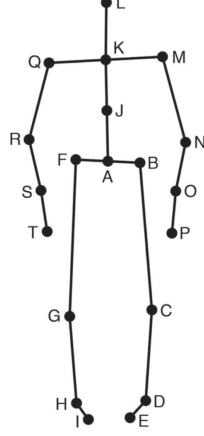


Figure 1: A Stick figure showing position of 20 markers

- **H3:** Synchronization between the joints in the upper body is more important for predicting interaction than the lower body.

Methods

We used a dataset of 128 motion capture recordings from [2] involving eight dyads dancing freely to 16 distinct musical stimuli. Each recording included 3D motion capture data from 20 markers (Figure 1). Mean interaction ratings were collected for each recording through a separate perceptual task.

We extracted Volumetric Matching, Torso Orientation, Imitation, and Synchrony features as outlined in the literature [3, 2]. In addition, we introduced two novel features, Energy and Covariance, which are derived from marker velocity. The velocity of each marker is computed using the Mocap toolbox from the position data. The dancer’s energy is defined as the mean square velocity across markers and time frames. Consequently, the dyad’s energy is the sum of the two dancers’ energies. The covariance of velocity between markers of two dancers x_i and x_j across X, Y, and Z dimensions is measured using correntropy—a non-linear covariance measure spanning the stimulus period.

$$K(x_i, x_j) = e^{\frac{-\|x_i - x_j\|_2^2}{2\sigma^2 T^2}} \quad (1)$$

Here, $\sigma=12(\text{constant})$ and T is the number of frames. With 60 time series for each dancer, the

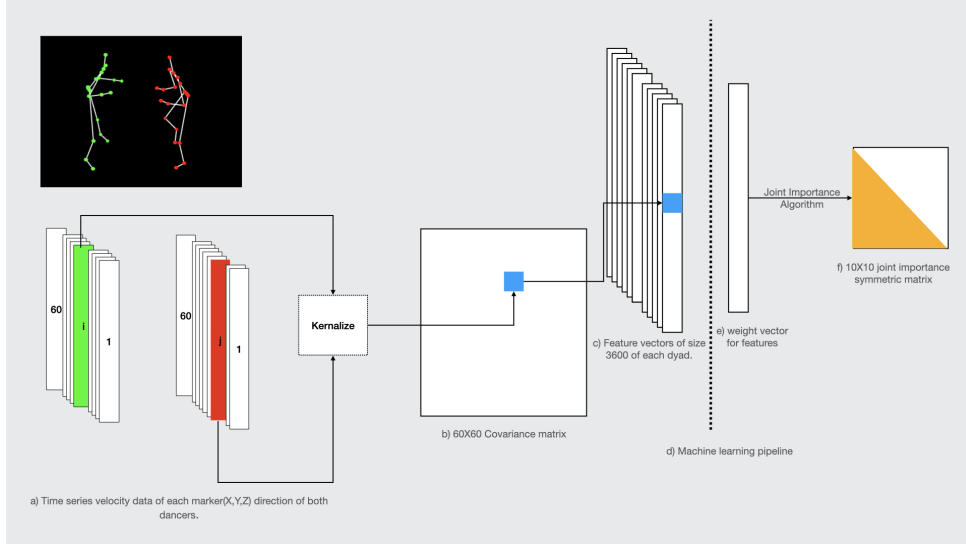


Figure 2: An Overview of the Machine Learning Pipeline for Extracting Key Joint Pairs in Interaction Prediction.

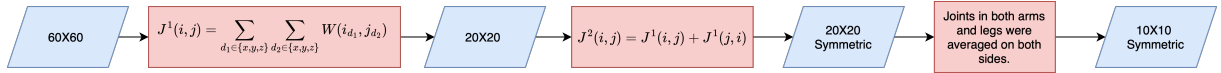


Figure 3: Joint Importance Algorithm

process yielded a 60x60 covariance matrix, resulting in a 3600-length feature vector.

We employed machine learning to predict interaction. Initial steps included categorizing interaction based on quantiles: 'low' ($< 1/3$ quantile), 'medium' ($1/3$ to $2/3$ quantile), and 'high' ($> 2/3$ quantile). For dimensionality reduction, Principal Component Analysis was applied, and Linear Support Vector Machine algorithm was employed for classification. We used nested cross-validation with stratified sampling to ensure generalizability, tuning hyperparameters like component retention and 'C' for SVM.

The machine learning model was trained on the covariance feature alone to identify significant joint pairs between dancers for predicting interaction. The entire pipeline is illustrated in (Figure 2). Using weights from the linear SVM, we identified the highest-weighted principal component and used its loadings' absolute values to measure marker pair importance. Using the series of transformations in Figure 3, the original 60x60 weight matrix is converted into a 10x10 symmetric matrix representing joint pair importance. Additionally, we evaluated the efficacy of features from literature along with our novel feature, Energy for predicting interaction. We trained the classifier separately on these features and then combined the features with the covariance matrix. We also explored two-class classifications using low and high categories.

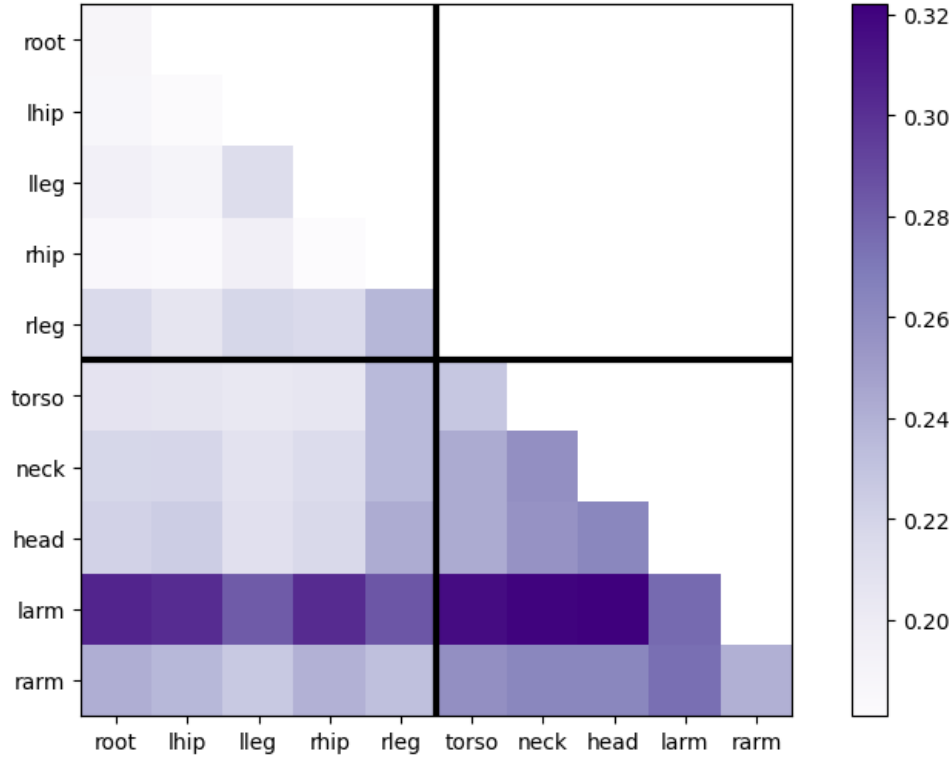


Figure 4: Joint Importance Matrix

	Two Class	Three Class
Covariance	0.72	0.41
Features	0.65	0.46
Combined	0.74	0.40

Table 1: 4-fold mean cross validated accuracies

Results

We found support for both hypotheses with a correlation of $r=0.38$ ($p<0.001$) for the first and $r=-0.33$ ($p<0.001$) for the second hypothesis. 4-fold mean cross-validation classification accuracies are reported in Table 1. The joint pair importance matrix is shown in Figure 4.

Discussion

As hypothesized, dyads with higher energy levels and similar dancer energies are perceived as more interactive. Accuracies exceeding chance were attained in both two-class and three-class classifications using all three feature sets. Furthermore, 72% accuracy using the covariance matrix on a two-class classification problem suggests we can rely on the model’s learned joint

pair importance weights to identify crucial joint pairs. As depicted in Figure 4, we can partition the matrix into three distinct quadrants corresponding to joint pairs in the lower body, between the upper and lower body, and the upper body. The figure demonstrates the importance of the upper body, confirming our third hypothesis. Consistent with earlier research involving dyads and other modalities such as spoken communication, the significance of arms becomes evident, as indicated by the prominent dark band. In conclusion, in addition to supporting our initial hypotheses, this study sheds light on the intricate dynamics of dyadic interaction and the pivotal role of the upper body, particularly the arms.

References

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